

SOUTHEAST UNIVERSITY

Department of Computer Science & Engineering



Home Fix

A Web-Based Home Service Management System

Project Members

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1. Project Goal :

Home Fix is a web-based Home Service Management System developed to simplify the process of booking and managing home maintenance services. The system allows customers to register, log in, browse available services, and submit service requests online. Customers can book services such as Electrician, Plumber, AC Repair, Carpenter, and Home Cleaning by providing their address, phone number, preferred service date, and problem description. The administrator manages customers, service providers, service categories, and booking requests. After receiving a booking, the administrator assigns an available service provider and updates the booking status from Pending to Assigned and finally Completed after the service is finished. Once a service is marked Completed, the customer can make a secure online payment for the service through the platform, and the administrator can view and manage the payment records. The system provides a centralized platform that improves service management, reduces manual work, and enables customers to track the status and history of their bookings.

2. System Features

Customer Features

- Register
- Login
- View Available Services
- Book a Service
- Fill Booking Form
 - Address
 - Phone Number
 - Preferred Service Date
 - Problem Description
- View Booking Status (Pending, Assigned, Completed)
- Make Payment for a Completed Service (via online payment gateway)
- View Payment Status and Receipt
- View Booking History

Admin Features

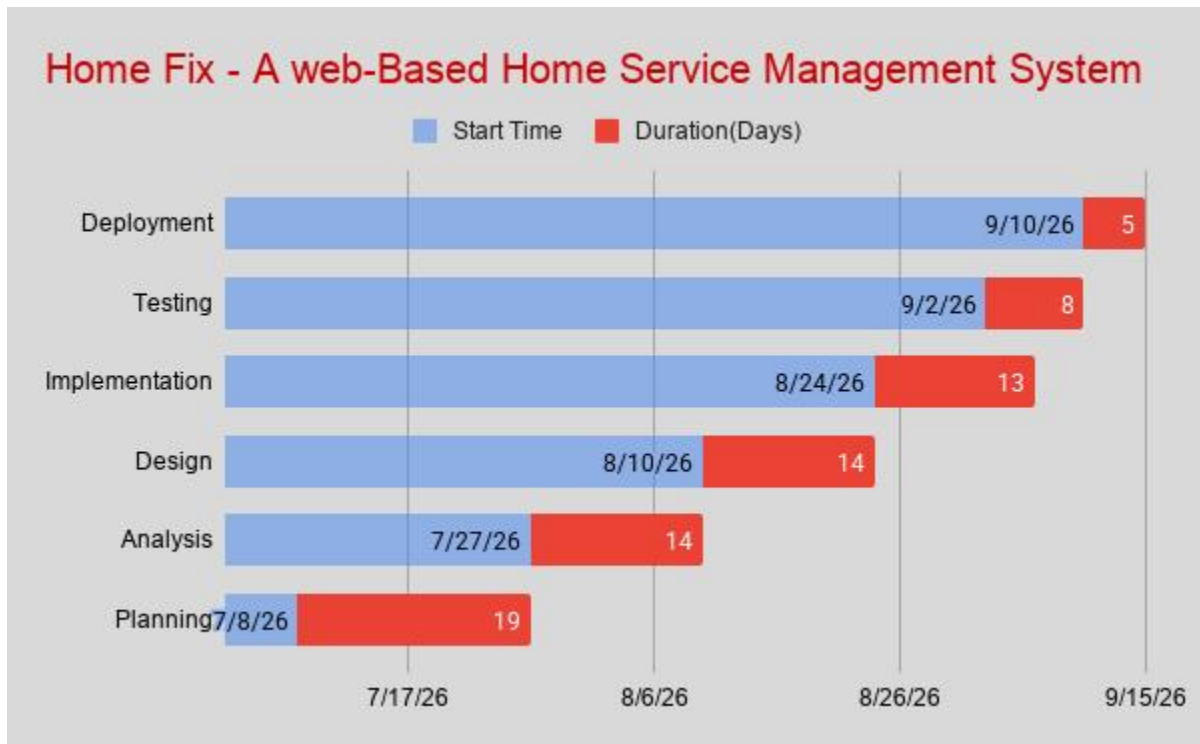
- Login
- Manage Customers
- Manage Service Providers (Add, Update, Delete)
- Manage Service Categories
- View All Bookings
- View Booking Details
- Assign a Service Provider to a Booking

- View Customer Details
- View Assigned Bookings
- Update Booking Status (Pending → Assigned → Completed)
- Mark Service as Completed
- View and Manage Payment Records
- Generate Payment Reports

3. Gantt Chart Summery:

Tasks	Start Time	Finish Time	Duration (days)
Planning	7/8/26	7/27/26	19
Analysis	7/27/26	8/10/26	14
Design	8/10/26	8/24/26	14
Implementation	8/24/26	9/6/26	13
Testing	9/2/26	9/10/26	8
Deployment	9/10/26	9/15/26	5

4.Gantt Chart:



5.Budget Preparation:

A. Personnel Costs :

Role	Description	Time	Budget (BDT)
Project Manager	Project planning, coordination and monitoring	73 Days (<i>full project</i>)	73,000
System Analyst	Requirements gathering and system analysis	14Days (<i>Analysis Phase</i>)	12,600
UI/UX Designer	User interface and user experience design	14 Days (<i>Design Phase</i>)	11,200
Frontend Developer (1)	Web application development	13 Days (<i>Implementation Phase</i>)	11,050
Backend Developer (2)	API development, business logic	13Days (<i>Implementation Phase</i>)	23,400
Database Admin	Database Design, Optimization	27 Days (<i>Design & Implementation</i>)	20,250
QA / Test Engineer	Software quality evaluation, debugging	8 Days (<i>Testing Phase</i>)	6,400
DevOps Engineer	Deployment, cloud configuration	5 Days (<i>Deployment Phase</i>)	4,250
Subtotal			162,150

B.Infrastructure & Software Cost

Item	Description	Time	Budget(BDT)
Cloud Hosting & Server	Application and database hosting for development through deployment	Project Duration & Initial Setup	8,000
Domain & SSL Certificate	Domain registration and SSL security	Annual / Setup	2,000
Development Tool	Project Management and Testing Tools	Project Duration	3,000

Database Management Software	Database Service and Backup Tools	Project Duration	3,000
Security Tools	Firewalls, Monitoring, Vulnerability Checking	Setup & Configuration	2,000
		Subtotal	18,000

C. Testing & Deployment Cost

Criteria	Description	Time	Cost(BDT)
Performance Testing	Load and performance testing across modules	8 Days (Testing Phase)	5,000
System & Security Testing	Security review and product deployment testing	5 Days (Deployment Phase)	4,000
Documentation & User Training	Technical and user documentation	5 Days (Deployment Phase)	3,000
		Subtotal	12,000

D.Maintenance Cost

Criteria	Description	Duration	Cost (BDT)
Maintenance Cost	Post-deployment maintenance and support	6 Months	20,000
Contingency	Reserve for emergency/unplanned costs	-	8,000
		Subtotal	28,000

E.Budget Summary

Criteria	Subtotal Cost (USD)	%
Personnel Cost	162,150	73.65%
Infrastructure & Software Cost	18,000	8.18%
Testing & Deployment Cost	12,000	5.45%
Maintenance Cost	28,000	12.72%
Total Cost	220,150	100%

6.Requirement Specification:

A.System Requirements:

Hardware Requirements

- Client-Side Devices: Customers and administrators can access HomeFix from any device with a modern web browser and an internet connection — desktop computers, laptops, tablets, or smartphones.
- Screen Resolution: Minimum 1024×768, with a fully responsive layout for smaller mobile screens.
- Development/Server Machine: Minimum 4 GB RAM, dual-core processor, and stable internet connectivity for development, testing, and hosting.

Software Requirements

- Frontend: HTML5, CSS3, and JavaScript (Bootstrap for a responsive, mobile-friendly UI).
- Backend: A server-side framework (e.g., PHP/Laravel or Node.js/Express) to handle business logic, authentication, and API endpoints.
- Database: MySQL for storing customer, service provider, service category, and booking data.
- Payment Gateway: A secure online payment gateway (e.g., SSLCommerz, bKash, or Stripe) integrated for processing customer payments.
- Hosting Platform: A cloud or shared hosting environment supporting the chosen backend and database (e.g., cPanel hosting or a cloud VPS).
- Browser Support: Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

B.User Requirements:

For Administrator

- Log in securely to access the Admin Panel.
- Manage customers, service providers, and service categories.
- View all incoming booking requests along with full booking details.
- Assign an available service provider to a pending booking.
- Update booking status from Pending to Assigned and finally to Completed.
- Mark a service as completed once finished.
- View customer details and their booking history.

For Customers

- Register for a new account and log in securely.
- Browse the list of available home services (Electrician, Plumber, AC Repair, Carpenter, Home Cleaning).
- Book a service by submitting address, phone number, preferred service date, and problem description.
- Track the real-time status of submitted bookings (Pending, Assigned, Completed).
- View their complete booking history.

C. Functional Requirements

- FR1:** The system shall allow customers to register and log in using secure credentials.
- FR2:** The system shall display all available service categories to customers, including Electrician, Plumber, AC Repair, Carpenter, and Home Cleaning.
- FR3:** The system shall allow customers to submit a booking request with address, phone number, preferred service date, and problem description.
- FR4:** The system shall allow administrators to log in and access a dedicated Admin Panel.
- FR5:** The system shall allow administrators to view, manage, and update customer and service-provider records.
- FR6:** The system shall allow administrators to assign an available service provider to a pending booking.
- FR7:** The system shall allow administrators to update a booking's status from Pending to Assigned and finally to Completed.
- FR8:** The system shall allow customers to view the current status and complete history of their bookings.
- FR9:** The system shall allow administrators to manage (add, update, delete) service provider records and service categories.
- FR10:** The system shall allow customers to make a secure online payment for a completed service and allow administrators to view and manage payment records.

D. Non-Functional Requirements

- **Performance:** Pages should load within 3 seconds under a standard broadband connection.
- **Scalability:** The system should support a growing number of customers, service providers, and simultaneous bookings without performance degradation.
- **Security:** Passwords must be securely hashed, all admin routes must be protected against unauthorized access, and all payment transactions must be processed through a PCI-compliant payment gateway using encrypted (HTTPS) connections.
- **Usability:** The interface must be clean, intuitive, and fully responsive across desktop, tablet, and mobile screens.
- **Reliability:** The system should maintain high uptime so customers can book services at any time.
- **Maintainability:** The codebase should be modular, well-commented, and version-controlled for easy future updates.
- **Data Storage & Backup:** All booking, customer, and provider data must be stored reliably in the database with regular backups.